

Total No. of Questions : 10]

SEAT No. :

P3322

[Total No. of Pages : 4

[5353]-197

T.E. (IT) (Semester - II)
SYSTEMS PROGRAMMING
(2012 Pattern)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume Suitable data if necessary.*

Q1) a) Give the various data structures in the design of 2-pass macroprocessor. **[6]**

b) Define the following : **[4]**

- i) Assembler
- ii) Macroprocessor
- iii) Compiler
- iv) Loader/Linker

OR

Q2) a) With the structure explain the different tables that would be generated as output of lexical analysis. **[4]**

b) What are the assembler directives? Explain how assembler directives LTORG, ORIGIN, START, END and EQU are processed with examples. **[6]**

Q3) a) For the following piece of assembly language code, show the contents of symbol table, literal table and pool-tab. Assume size of instruction equal to one. **[6]**

START 500

MULT BREG, A

P.T.O.

```

MOVEM AREG , = '10'
LOOP MOVER AREG, A
    MOVER CREG,B
    ADD CREG, = '1'
    SUB CREG,A
    LTORG
    ADD CREG,B'
NEXT SUB AREG='1'
STOP
ORIGIN 300
MULT CREG, B
ADS 1
BACK EQU LOOP
B DS 1
END

```

- b) Define loader and enlist the basic functions of loader. [4]

OR

- Q4)** a) Explain the phases of compiler w.r.t the following statement : [8]

$$R = (b*b - 4*a*c)/(2*a)$$

- b) Define the term forward reference in an assembler. [2]

- Q5)** a) Consider the grammar [8]

$E \rightarrow E-E$

$E \rightarrow E * E$

$E \rightarrow id$

Perform shift Reduce parsing of i/p string "id – id * id"

- b) Explain LEX file structure. [6]

- c) Compare bottom up and top down parser. [4]

OR

Q6) a) Consider the following grammar

$S \rightarrow iEtSS'/a$

$S' \rightarrow eS/e$

$E \rightarrow b$

Design a table driven predictive parser and parse the string 'ibtae'. [8]

b) Explain YACC file structure. [5]

c) With a neat diagram explain the classification of parsers. [5]

Q7) a) Define synthesized and inherited attributes. For the grammar given, [8]

$T \rightarrow FT'$

$T' \rightarrow *FT'/\epsilon$

$F \rightarrow \text{digit}$

Draw the annotated parse tree for the expression $3*5*2$ and list down the synthesized and inherited attributes.

b) Write the following expression in the form of postfix notation, Directed acyclic graph, quadruple and triple. [8]

$a = b * (-c) + b * (-c)$

OR

Q8) a) Draw the dependency graph for the example of 7 a. [4]

b) Translate the following C fragment into the three address code. [6]

int i,j,k;

int T[5][100];

j = 10; k = j-1;

for (i = 1; i <= 5; i++)

{

 T[i][j]=k*k;

 J=j+k

 T[i][j]=k*k-1;

}

c) What is the need for intermediate code generation?

Explain the different IC forms with examples. [6]

Q9) a) Optimize the following code **[8]**

```
i=1
j=1
t1=i * 80
t2=j*4
t3=t1+t2
t4=a[t3]
t5=j*80
t6=i*4
t7=t5+t6
t8=a[t7]
```

b) Write short note on activation record. **[4]**

c) Explain in brief run time storage allocation. **[4]**

OR

Q10)a) Generate three address code for **[8]**

```
while (a<c) and (b>d) do
{
  If a =1 then c=c+1
  Else while (a<=d) do
    A=a+3
}
```

b) Explain different storage allocation strategies. **[4]**

c) Explain various code generation issues. **[4]**

